Phase 3: Requirements Gathering

Project: BloodLink | رابط الحياة
Program: DEPI (Digital Egypt Pioneers Initiative), 2026
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Contents

1. Stakeholder Analysis
2. User Stories
3. Use Cases
4. Functional Requirements
5. Non-functional Requirements

Priority: Must is in the core and done by checkpoint 2 (13 Nov). **Should** is a planned extra for sprint 5. **Could** comes after the course.

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1. Stakeholder Analysis

Stakeholder	Who they are	What they need	Influence	How we involve them
Requester	A patient's family member or friend	Find compatible donors fast, without posting everywhere	High	Interviews, usability test of the request flow
Donor	A volunteer who wants to give blood	Only relevant, real requests close to home; privacy	High	Short survey, usability test of the donor flow
Partner hospital	Hospital staff using our portal	Confirm papers quickly, record donations, reach donors for their patients	High	Talk to 1 or 2 hospitals; demo uses fictional ones
Non-partner hospital	Any other official hospital	Nothing from us; it still examines the donor and decides	Low	None. Its papers are checked by AI
Admin	Our moderation team	Review unclear papers, handle reports, ban abusers	Medium	Built by the team
AI service provider	The vision AI API	Clear prompts, sample papers only on the free tier	Low	Through one `checkPaper()` function
DEPI lecturer and evaluators	Grade the project	Clear docs, working demo, good code	High	Phase reviews and the final presentation
Project team	The 6 members	A realistic plan and clear roles	High	Weekly meetings, GitHub board

2. User Stories

Format: *As a [role], I want [goal], so that [benefit].*

Donor

ID	Story	Acceptance criteria	Priority
US-01	As a donor, I want to sign up with my blood type and area, so that I only get requests that fit me	Signup asks for name, phone, blood type, governorate, area and location. The user must accept the terms of use.	Must
US-02	As a donor, I want to upload my hospital blood-type certificate, so that people can trust me	Accepts JPG, PNG or PDF up to 5 MB. Shows the resulting badge: **Verified**, **Paper looks valid** or **Needs review**.	Must
US-03	As a donor, I want to be notified about requests near me, so that I can help in time	A push notification arrives only when the blood type is compatible, I'm available, and the hospital is within the wave radius.	Must
US-04	As a donor, I want to accept or decline a request, so that the requester knows quickly	One tap to accept or decline. On accept, I see the hospital address and the requester's phone number.	Must
US-05	As a donor, I want a QR code when the hospital is a partner, so that the staff can record my donation	QR shows only for partner hospitals. At other hospitals, I see "No QR here, confirm after donating."	Must
US-06	As a donor, I want to turn my availability off, so that I'm not disturbed	A visible on/off switch. No notifications while it's off.	Must
US-07	As a donor, I want to see my donation history, so that I feel the impact	List of past donations with date, hospital and how each was confirmed	Should

Requester

ID	Story	Acceptance criteria	Priority
US-08	As a requester, I want to create a request with the hospital paper, so that donors trust it	The request can't be sent without a paper. Asks for blood type, units, hospital, urgency and deadline.	Must
US-09	As a requester, I want to see the badge my request got, so that I know how it looks to donors	Shows **Verified** (partner confirmed), **Paper looks valid** (AI) or **Needs review**	Must
US-10	As a requester, I want to follow the progress live, so that I don't keep calling people	"2 of 3 units pledged" updates without refreshing the page	Must
US-11	As a requester, I want to see a donor's phone number only after they accept, so that contact is safe for both sides	No phone numbers appear before acceptance	Must
US-12	As a requester at a non-partner hospital, I want to confirm the donation myself, so that the request can be closed	Both requester and donor confirm. The donor's badge does **not** change.	Must
US-13	As a requester, I want to report a suspicious donor or message, so that abusers are removed	Report button on every profile and request, with a reason	Must
US-14	As a requester, I want to share my request on WhatsApp, so that more people see it	Share card with a link back to the checked request	Should

Partner hospital

ID	Story	Acceptance criteria	Priority
US-15	As partner-hospital staff, I want to confirm papers issued by my hospital, so that requests and donors get the **Verified** badge	List of pending papers with Confirm and Reject buttons	Must
US-16	As partner-hospital staff, I want to scan a donor's QR code, so that I can record what happened	Outcome: **Donated**, **Not able to donate today (medical decision)** or **Didn't show up**	Must

Admin

ID	Story	Acceptance criteria	Priority
US-17	As an admin, I want to review papers the AI marked as unclear, so that no user is stuck	Queue of unclear papers with Approve, Ask for a clearer photo and Reject	Must
US-18	As an admin, I want to handle reports and ban accounts, so that the community stays safe	Reports list with Dismiss, Warn and Ban. Banned users can't log in.	Must
US-19	As an admin, I want to approve partner hospitals, so that only real hospitals can confirm papers	Partner accounts are inactive until an admin approves them	Must
US-20	As an admin, I want to see key numbers, so that I know if the platform works	Cards: active requests, donors, average time to first acceptance, open reports	Must

3. Use Cases

UC-01: Create a blood request

Item	Details
Actor	Requester
Precondition	Logged in, terms accepted
Main flow	1. Fill in blood type, units, hospital, urgency and deadline. 2. Upload the hospital paper. 3. Review the "check your answers" page. 4. Send. 5. The system checks the paper: partner hospital, so it goes to the hospital to confirm; non-partner, so the AI checks it. 6. The request becomes active with a badge and matching starts.
Alternative flows	5a. The AI finds the paper unclear: the request waits in the admin review queue. 5b. The partner hospital rejects it: the request is rejected and the requester is told why.
Postcondition	An active request with a trust badge, and the first wave of notifications sent

UC-02: Notify donors and accept

Item	Details
Actor	System, Donor
Precondition	An active request
Main flow	1. The system finds compatible, available donors within 5 km and ranks them (distance, same type, trust, reliability). 2. It sends push notifications. 3. A donor opens the request and taps **I can donate**. 4. The pledge count goes up live for the requester. 5. Phone numbers are shown to both.

Item	Details
Alternative flows	1a. Not enough donors accept: the next wave widens to 15 km. 3a. The donor declines: the system notifies the next donor. 3b. Nobody accepts before the deadline: the request expires.
Postcondition	A pledge linked to the request

UC-03: Record a donation at a partner hospital

Item	Details
Actor	Donor, Partner-hospital staff
Precondition	The donor accepted a request at a partner hospital
Main flow	1. The donor shows the QR code at the hospital. 2. Staff scan it in the portal. 3. The hospital examines the donor and decides. 4. Staff record **Donated**. 5. The donor becomes **Verified** and is paused from new requests for a while. 6. The request count updates.
Alternative flows	4a. **Not able to donate today:** the pledge is released and the next wave may start. 4b. **Didn't show up:** the donor's reliability score goes down.
Postcondition	A recorded donation, confirmed by the hospital

UC-04: Confirm a donation at a non-partner hospital

Item	Details
Actor	Donor, Requester
Precondition	The donor accepted a request at a non-partner hospital
Main flow	1. They meet at the hospital, which decides medically. 2. The donor taps **I donated**. 3. The requester taps **Confirm**. 4. The donation is recorded as self-confirmed.
Alternative flows	3a. The requester says it didn't happen: the case goes to an admin.
Postcondition	A self-confirmed donation. The donor's badge stays the same.

UC-05: Handle a report

Item	Details
Actor	Any user, Admin
Main flow	1. A user reports a profile or request with a reason, for example asking for money. 2. The admin reviews it. 3. The admin dismisses it, warns the user or bans the account. 4. The reporter is told the result.
Postcondition	The report is closed and the action is logged

4. Functional Requirements

ID	Requirement	Priority
Accounts		
FR-01	Users can register and log in with email or phone and a password	Must

ID	Requirement	Priority
FR-02	The system supports four roles: donor, requester, partner hospital, admin. A user can be both donor and requester.	Must
FR-03	Users must accept the terms of use (the site only connects people) at signup	Must
FR-04	Partner-hospital accounts stay inactive until an admin approves them	Must
Papers and trust		
FR-05	Donors upload an official hospital blood-type certificate	Must
FR-06	Requesters must upload an official hospital paper before a request is published	Must
FR-07	Papers from partner hospitals are sent to that hospital to confirm, and give the **Verified** badge	Must
FR-08	Papers from other hospitals are checked by AI, and give the **Paper looks valid** badge	Must
FR-09	The AI check confirms the paper looks official, the name and blood type match, and the date is recent. Unclear results go to the admin queue.	Must
FR-10	The AI never labels a paper as genuine. The badge text is fixed.	Must
FR-11	Uploaded papers are private and never shown to other users	Must
Matching and notifications		
FR-12	Matching uses blood compatibility, availability, pause period, distance and trust level	Must
FR-13	Notifications go out in waves: 5 km first, then 15 km	Must
FR-14	Donors get browser push notifications and a list of requests on the website	Must
FR-15	Requests expire automatically after their deadline	Must
Response and meeting		
FR-16	Donors can accept or decline a request	Must
FR-17	Phone numbers are shown only after the donor accepts	Must
FR-18	Requesters see the pledge count live	Must
FR-19	At partner hospitals, the donor gets a QR code, and staff scan it to record **Donated**, **Not able to donate today** or **Didn't show up**	Must
FR-20	At non-partner hospitals, there is no QR code. Donor and requester confirm the donation themselves. This never upgrades the donor's badge.	Must
FR-21	After a donation, the donor is paused from new requests for a period (general guidance; the hospital decides)	Must
Safety and admin		
FR-22	Every profile and request has a report button	Must
FR-23	Admins can review unclear papers, handle reports and ban accounts	Must
FR-24	The admin dashboard shows key numbers: active requests, donors, time to first acceptance, open reports	Must
Extras		
FR-25	English version of the interface	Should

ID	Requirement	Priority
FR-26	Map view of requests	Should
FR-27	WhatsApp share card	Should
FR-28	Donation history and charts	Should
FR-29	SMS alerts, blood stock at partner hospitals	Could

5. Non-functional Requirements

ID	Category	Requirement	How we check it
NFR-01	Performance	95 % of API requests answer in under 500 ms	Server logs
NFR-02	Performance	Pages load in under 3 seconds on a phone over 4G	Lighthouse
NFR-03	Performance	Matching for one request finishes in under 2 seconds	Unit test with demo data
NFR-04	Security	Passwords are hashed with bcrypt. Login uses a JWT in an httpOnly cookie.	Code review
NFR-05	Security	Every input is validated on the server (zod). Login and upload are rate-limited.	Code review, API tests
NFR-06	Security	No secrets in the repo; all keys live in `.env`	CI check, `.env.example`
NFR-07	Privacy	Phone numbers and papers are shared only as described in FR-11 and FR-17. Users consent before the AI check. We follow Egypt's Personal Data Protection Law (151/2020).	Code review, test cases
NFR-08	Privacy	Only sample papers are sent to the free AI tier	Team rule, checklist
NFR-09	Usability	Arabic first, right-to-left layout, works on phones from 360 px wide	Manual test on phones
NFR-10	Usability	A first-time requester can send a request in under 3 minutes	Usability test
NFR-11	Accessibility	Lighthouse accessibility score ≥ 90, visible keyboard focus, readable contrast	Lighthouse
NFR-12	Reliability	Uptime ≥ 99 % during demo week	Uptime monitor
NFR-13	Reliability	If the AI service fails, the website switches to mock mode or the review queue, never to an error page	Test with the AI turned off
NFR-14	Maintainability	TypeScript on both sides, ESLint and Prettier, small reviewed pull requests	CI
NFR-15	Testability	≥ 70 % test coverage of core logic (matching, trust badges, request states)	Jest coverage report
NFR-16	Portability	The whole website runs locally with `docker compose up`	Fresh-laptop test
NFR-17	Cost	The demo runs at $0 on free tiers	Monthly check of all accounts